

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M.Tech Examination (IT)

May 2018

IT745: Data Science & Modelling

Date: 03.05.2018, Thursday

Time: 01:30 pm to 04:30 pm

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

[7]

- 1 Can a data mining system find **all and only** interesting patterns? Justify your answer. 3
- 2 What is Data Mining? Differentiate between Descriptive and Predictive data mining functionality with example. 4

Q-2

[14]

- (a) Explain with example evolution of sciences from empirical science to data science. 4
 - (b) **Answer following questions.(Any Two)** 10
- 1 A random sample of 395 people were surveyed and each person was asked to report the highest education level they obtained. The data that resulted from the survey is summarized in the following table:

	High School	Bachelors	Masters	Ph. d.	Total
Female	60	54	46	41	201
Male	40	44	53	57	194
Total	100	98	99	98	395

Is gender independent of education level? The critical value of with 3 degree of freedom is 7.815. Find correlation between gender and education level using chi square method.

- 2 What are the advantages of sampling as a data reduction technique over other techniques? Explain in brief Sampling with replacement, Cluster Sampling and Stratified Sampling.
- 3 Why is data preprocessing important? What are the major tasks in data preprocessing? How to handle missing data?

Q-3 Attempt Any Two.

[14]

- (a) The following contingency table summarizes supermarket transaction data, where A refers to the transactions containing A, $\sim A$ refers to the transactions that do not contain A, B refers to the transactions containing B, and $\sim B$ refers to the transactions that do not contain B.

	A	$\sim A$
B	2000	500
$\sim B$	1000	1500

1. Suppose that the association rule " $A \Rightarrow B$ " is mined. Given a minimum support threshold of 25% and a minimum confidence threshold of 50%, is this association rule strong?
2. Based on the given data, is the purchase of 'A' independent of the purchase of 'B'? If not, what kind of correlation relationship exists between the two using lift as a correlation measure?

- (b) What is importance of normalization in data preprocessing? Use the methods below to normalize the following group of data:
20, 30, 40, 60, 100
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization
(c) normalization by decimal scaling
- (c) What are the three phases of Information Extraction? Explain in detail.

SECTION-II

- Q-4** [7]
1 What is the limitation of accuracy as a classifier performance measure? What other measures can be used in place of accuracy? 3
2 Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8): 4
(i) Compute the Euclidean distance between the two objects.
(ii) Compute the Manhattan distance between the two objects.
- Q-5** [14]
(a) A database has four transactions. Let *minimum support* = 50% and *minimum confidence* = 70%. 4

TID	Items
T1	A,C,D
T2	B,D
T3	A,B,C,E
T4	B,D,F

Find all frequent item sets using Apriori.

- (b) **Answer following Questions. (Any Two)** 10
1 What is over fitting in classification? How tree pruning is helpful in handling over fitting problem of classification?
2 Explain Convolution Neural Network (CNN) with example.
3 What is difference between classification and clustering? Compare the advantages and disadvantages of *eager* classification versus *lazy* classification.
- Q-6 Attempt Any Two.** [14]
(a) Suppose that the data mining task is to cluster the following seven points (with (x; y) representing location) into two clusters.
A1(1.0,1.0); A2(1.5,2.0); A3(3.0, 4.0); A4(5.0, 7.0); A5(3.5,5.0); A6(4.5, 5.0); A7(3.5,4.5) The distance function is Euclidean distance. Suppose initially we assign A1 and A4 as the center of each cluster, respectively. Use the k-means algorithm to find final cluster.
(b) What is significance of dimensionality reduction in data mining? Explain the problem of curse of dimensionality. Explain following dimensionality reduction techniques.
1. step-wise forward selection
2. step-wise backward elimination

- (c) What is the basic idea behind Naïve Bayes Classifier? The following table consists of training data from student database. The data have been generalized. Let buys computer be the class label attribute.

<i>age</i>	<i>income</i>	<i>student</i>	<i>credit_rating</i>	<i>Class: buys_computer</i>
youth	high	no	fair	no
youth	high	no	excellent	no
middle_aged	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle_aged	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle_aged	medium	no	excellent	yes
middle_aged	high	yes	fair	yes
senior	medium	no	excellent	no

Given a test tuple having the values $X = (\text{age} = \text{young}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$. What would naive Bayesian classification for the tuple be?